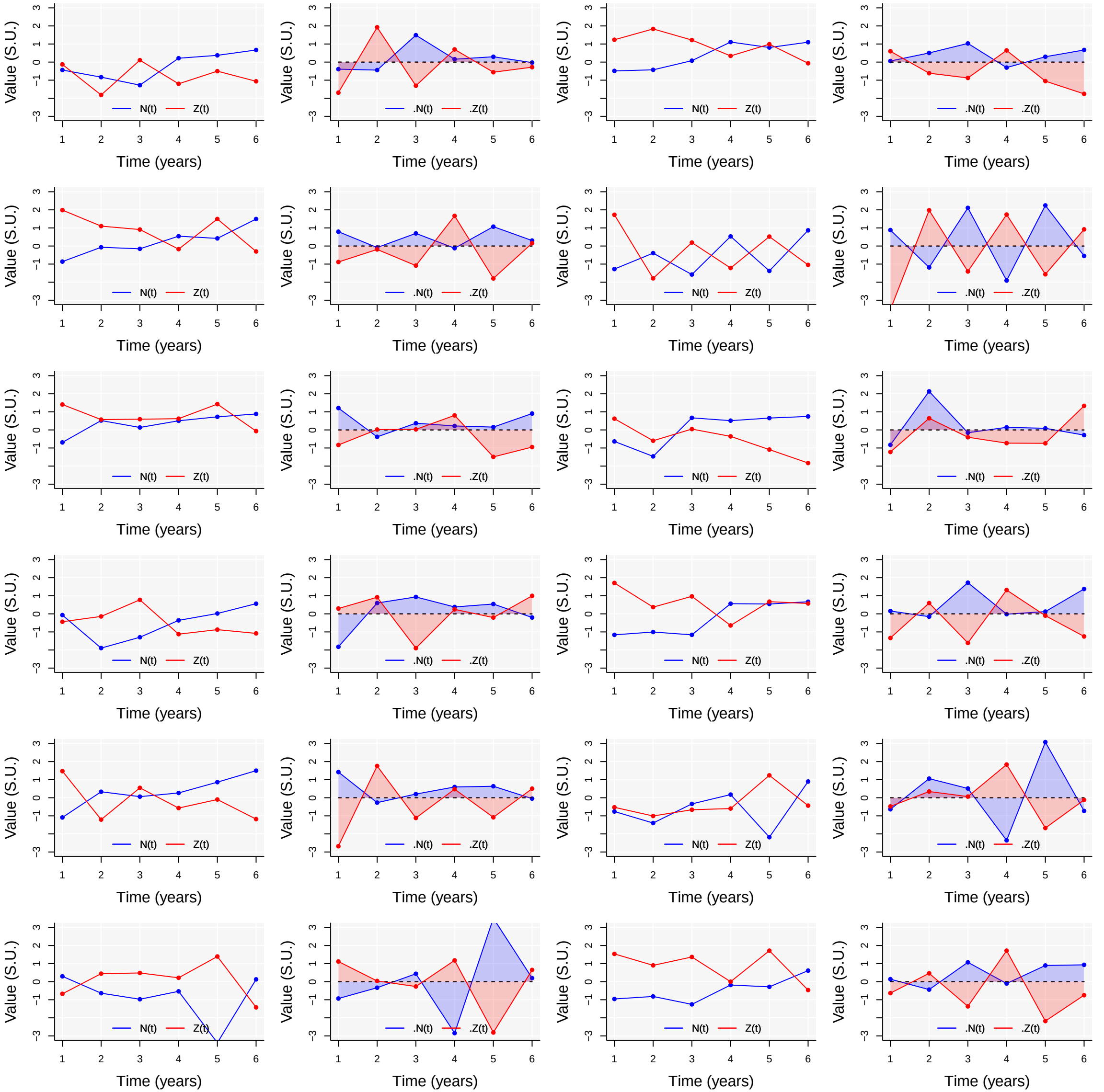
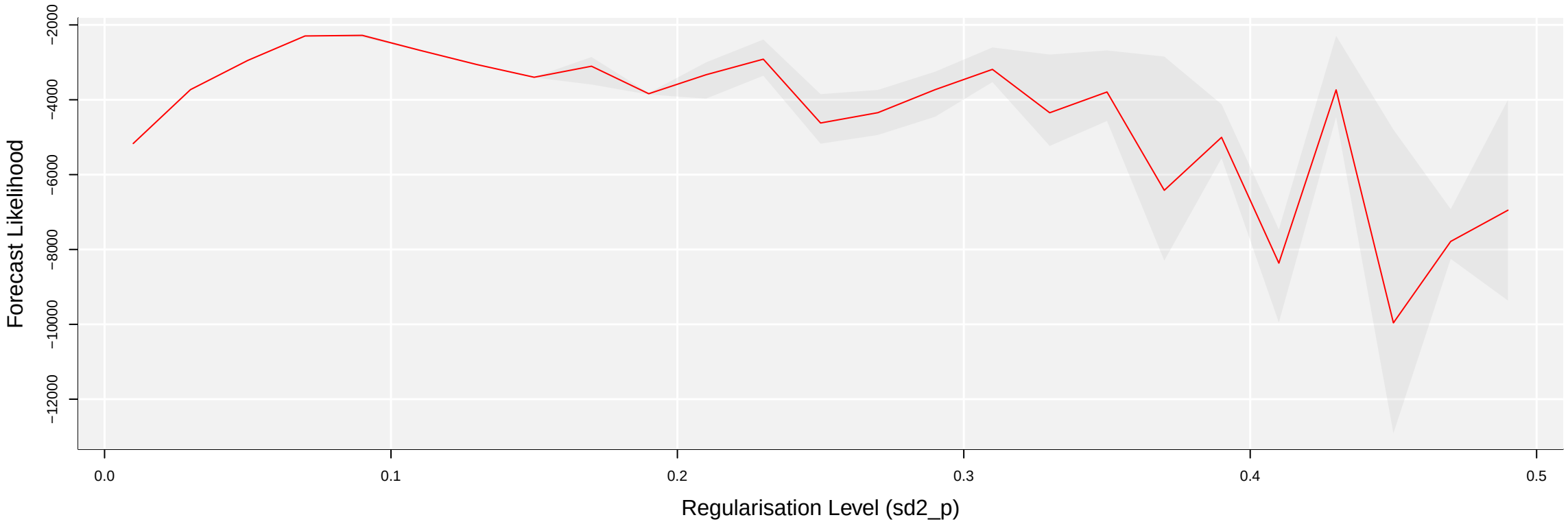
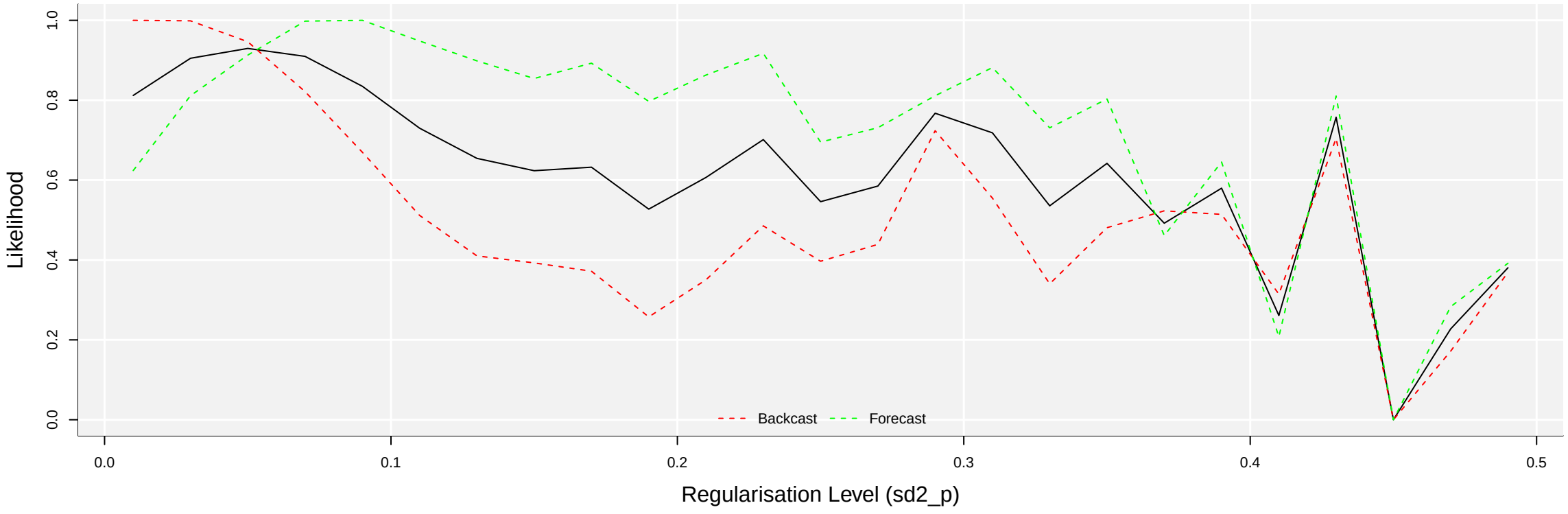
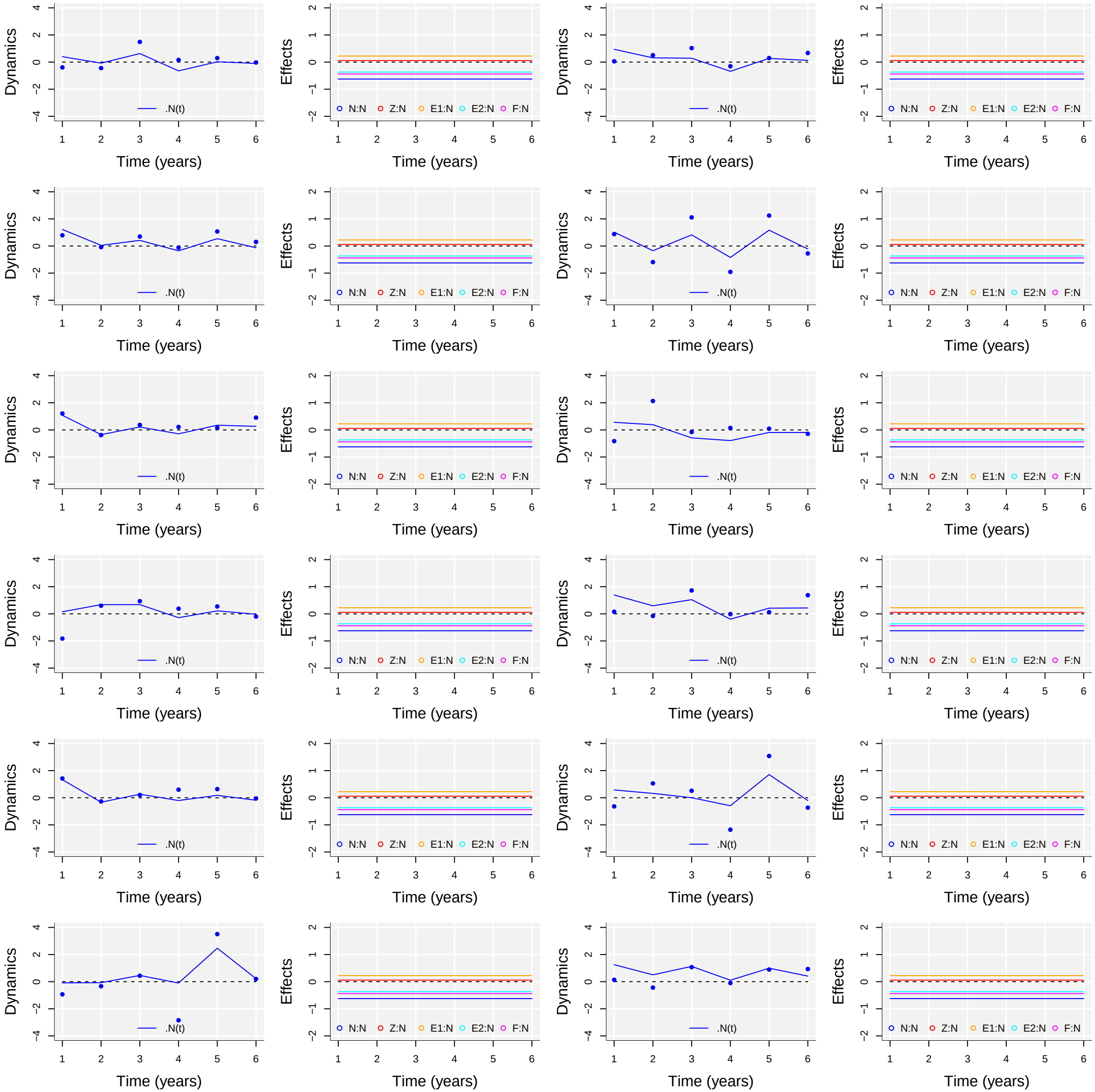
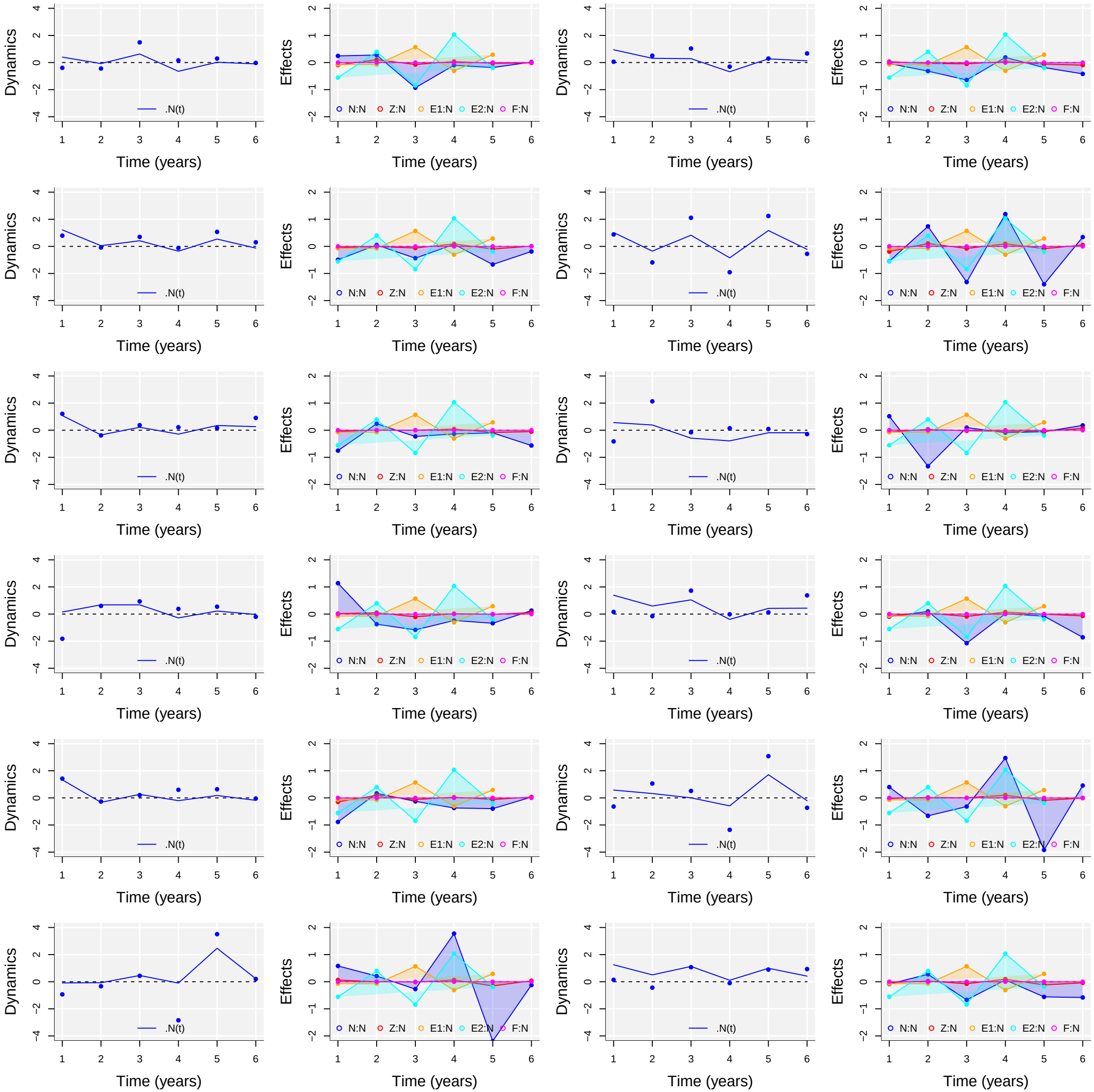

Value (S.U.)
Time (years)
N(t)
Z(t)
.N(t)
.Z(t)

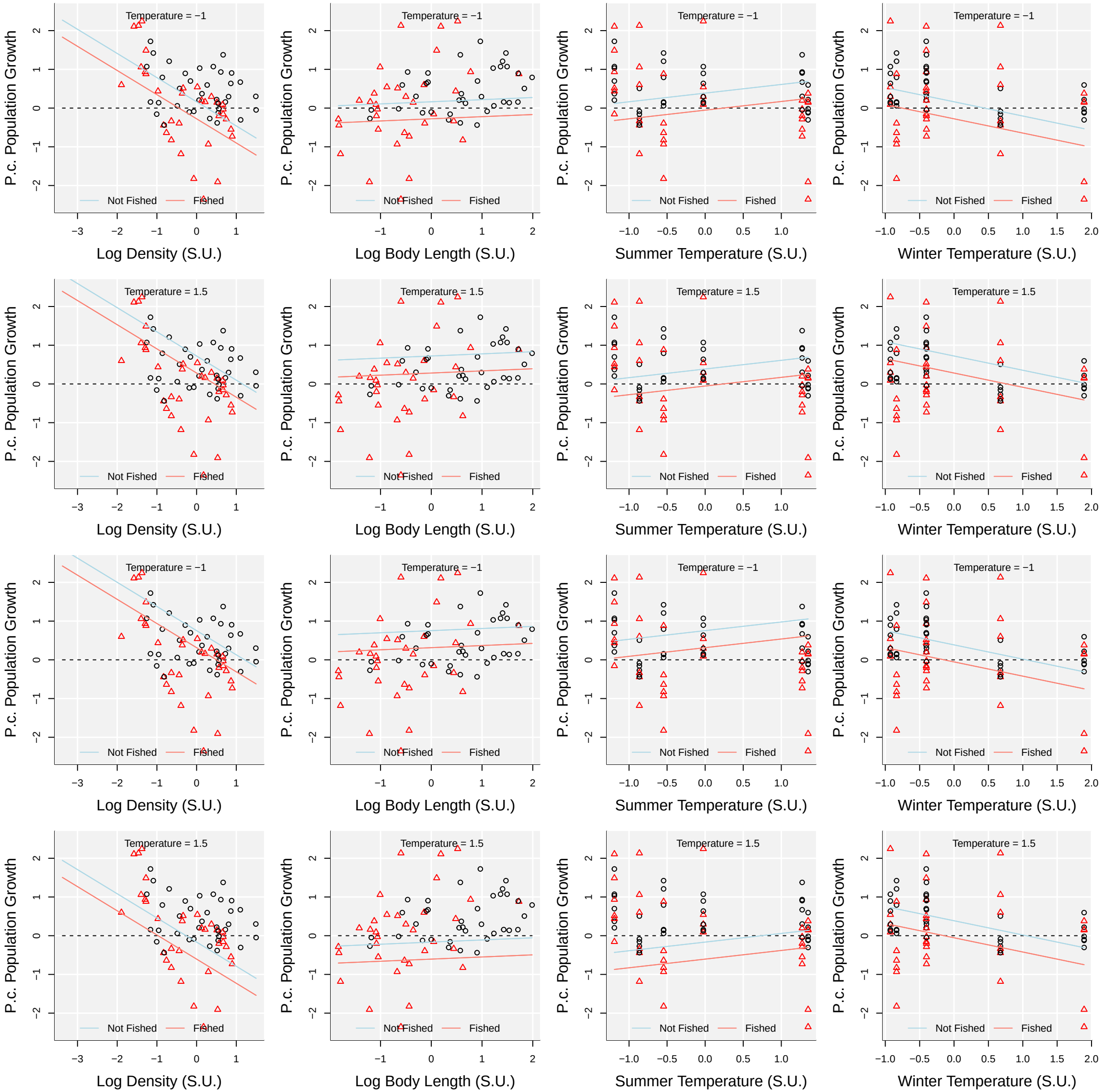
Backcast Likelihood
−2000
−3000
−4000
−5000
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

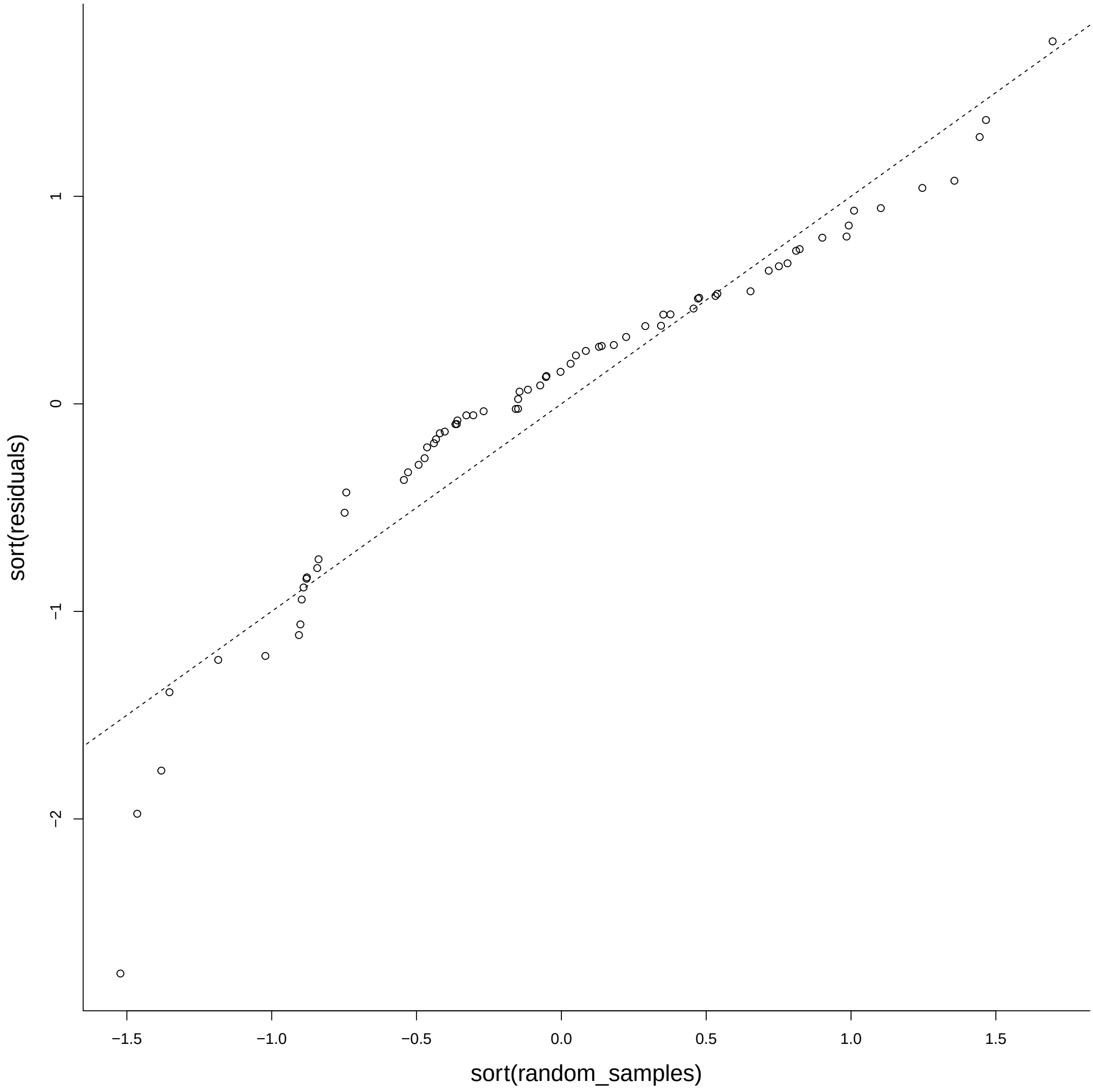

Forecast Likelihood
−2000
−4000
−6000
−8000
−10000
−12000
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

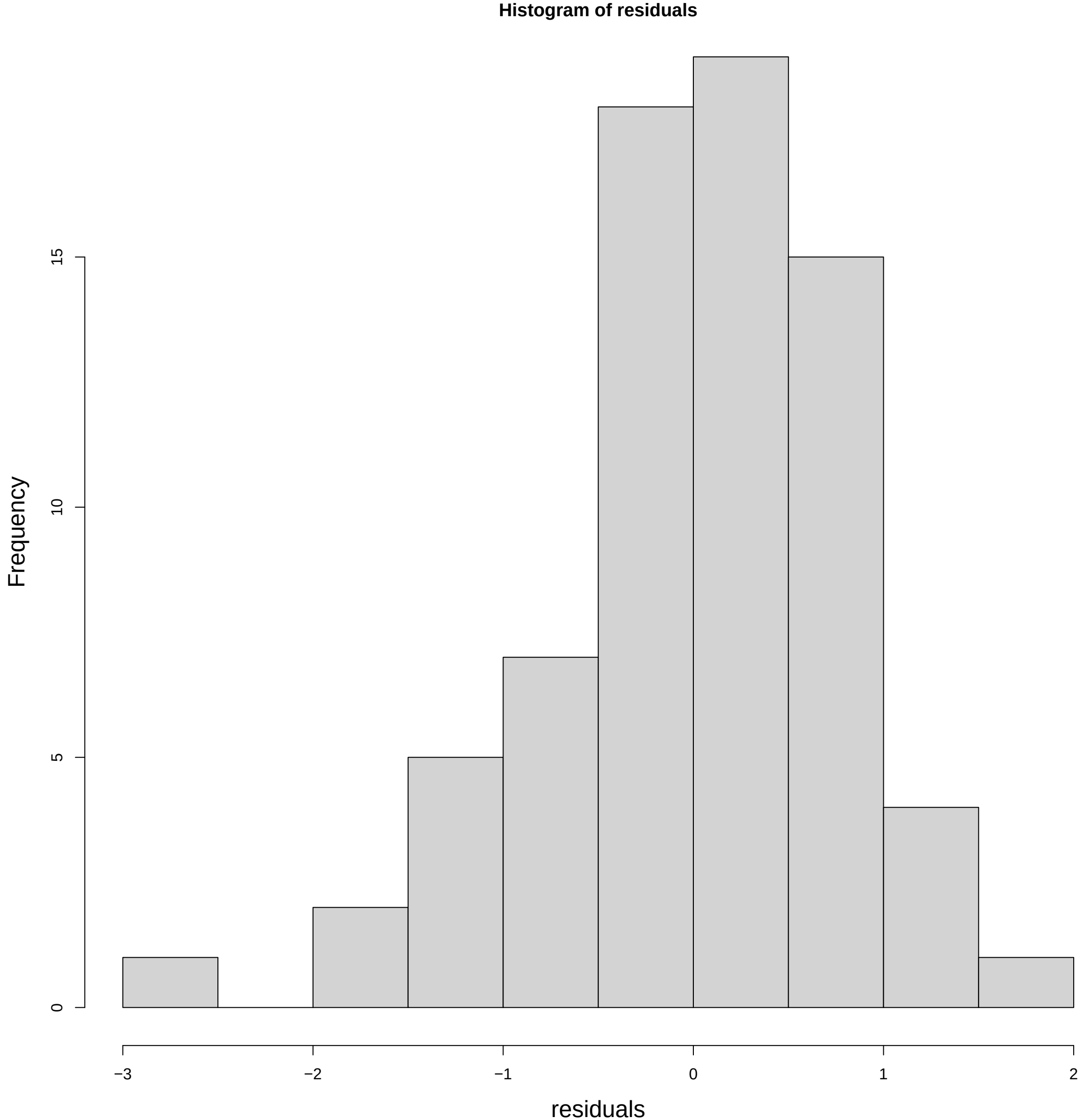

Likelihood
1.0
0.8
0.6
0.4
0.2
0.0
Backcast
Forecast
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

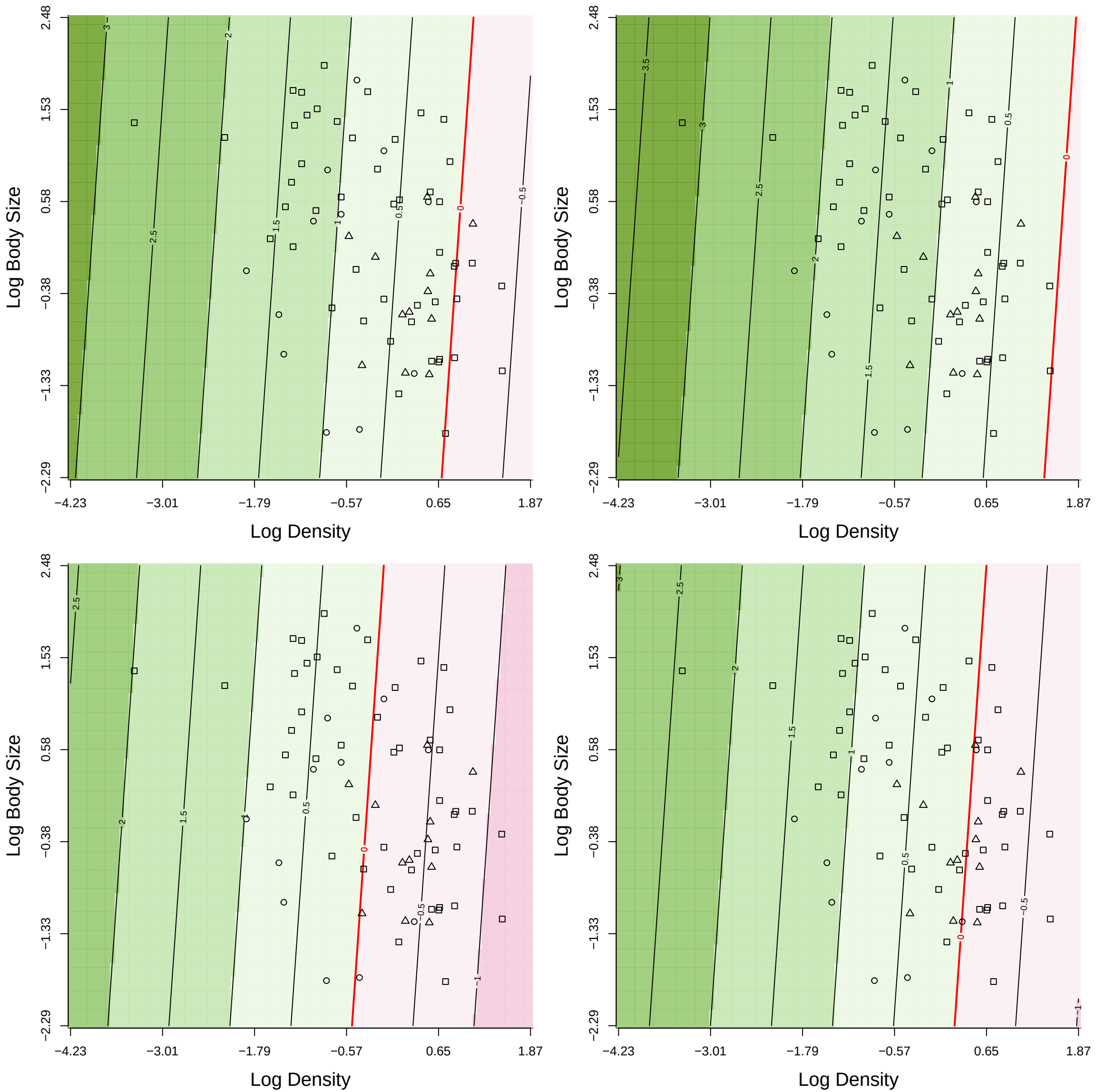

Dynamics
Effects
Time (years)
.N(t)
N:N
Z:N
E1:N
E2:N
F:N

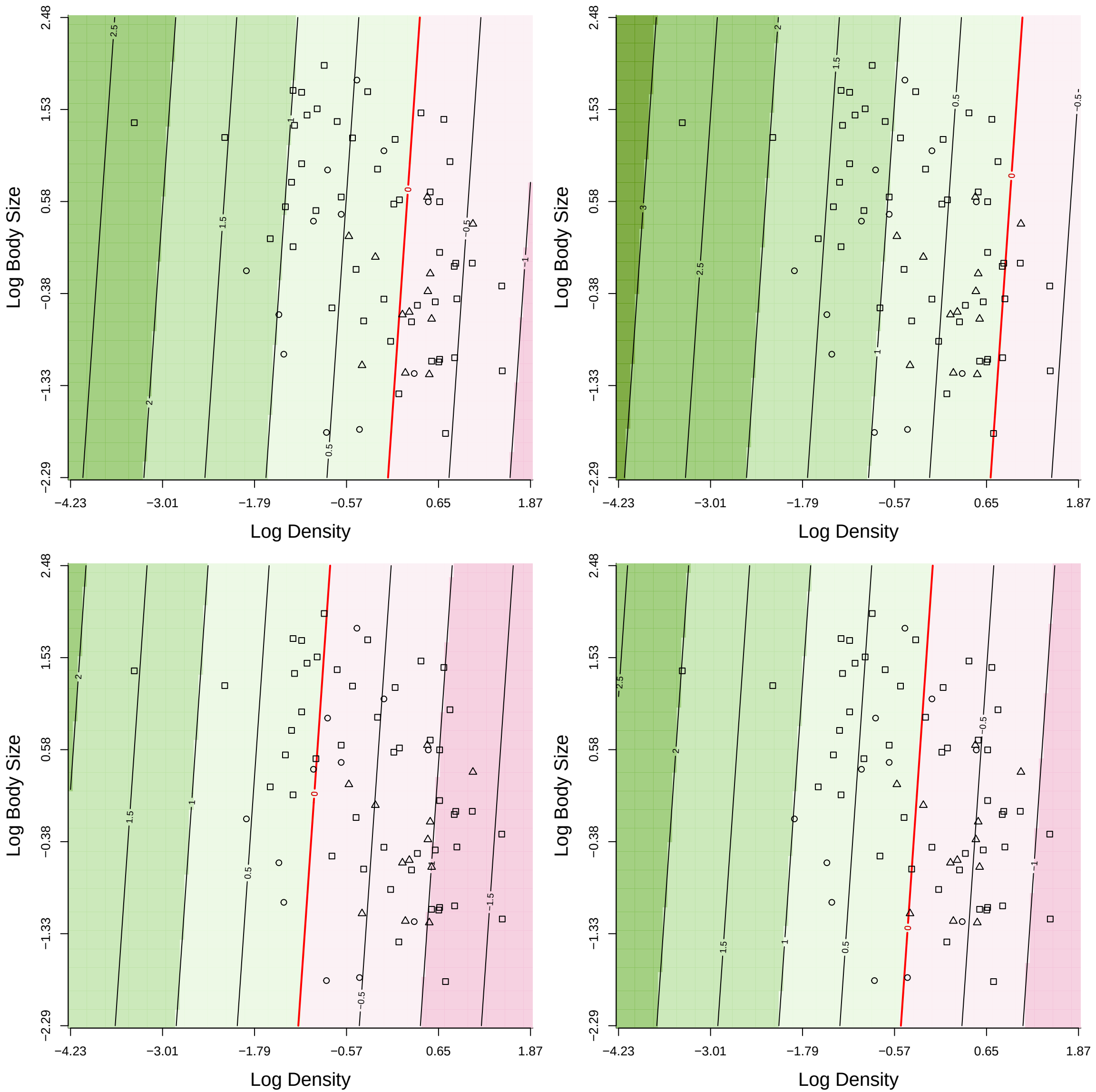

Dynamics
Effects
Time (years)
.N(t)
N:N
Z:N
E1:N
E2:N
F:N

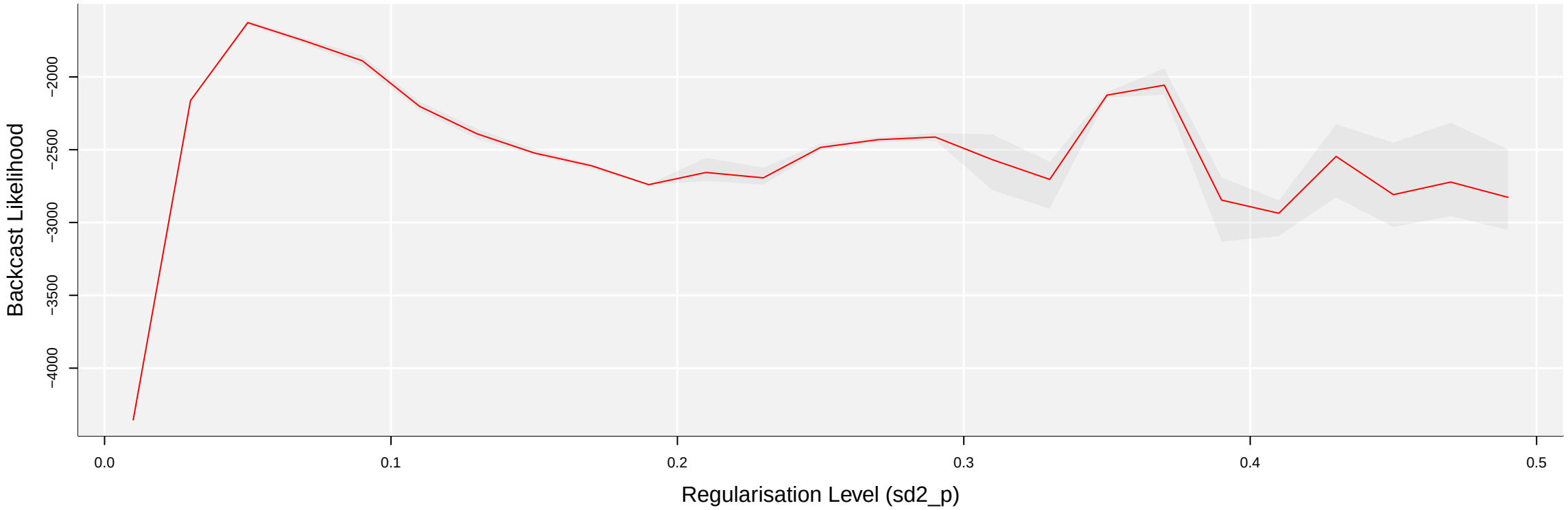

Temperature = −1
Temperature = 1.5
P.c. Population Growth
Log Density (S.U.)
Log Body Length (S.U.)
Summer Temperature (S.U.)
Winter Temperature (S.U.)
Not Fished
Fished

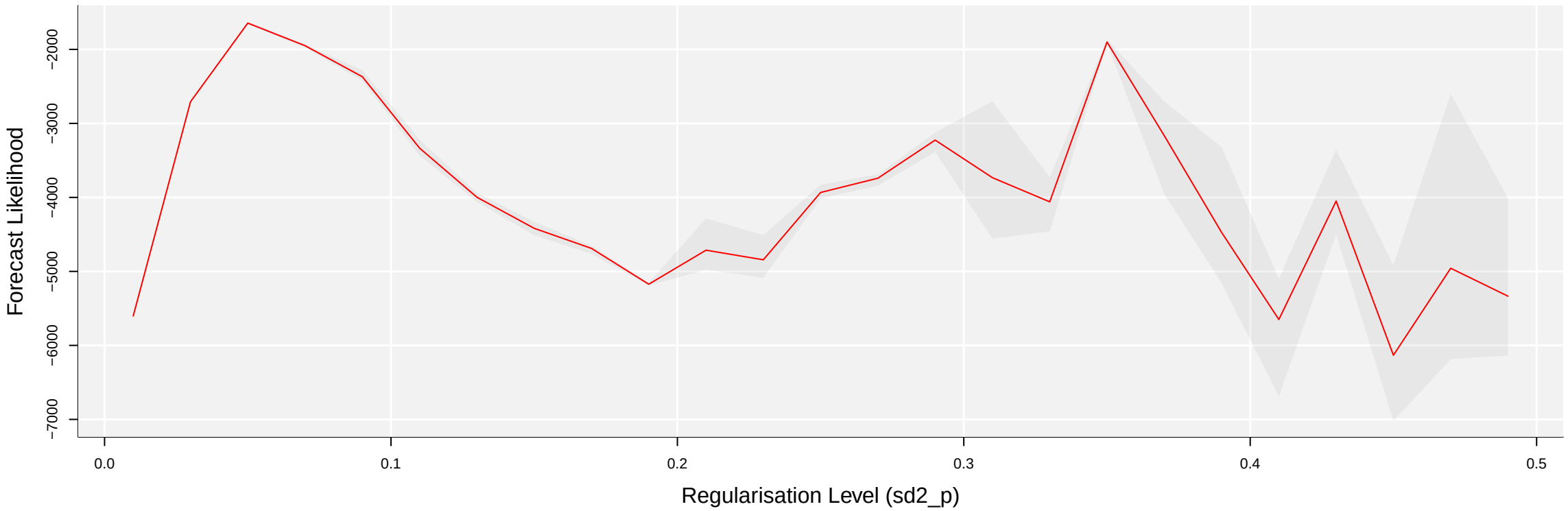

sort(residuals)
1
0
−1
−2
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
sort(random_samples)

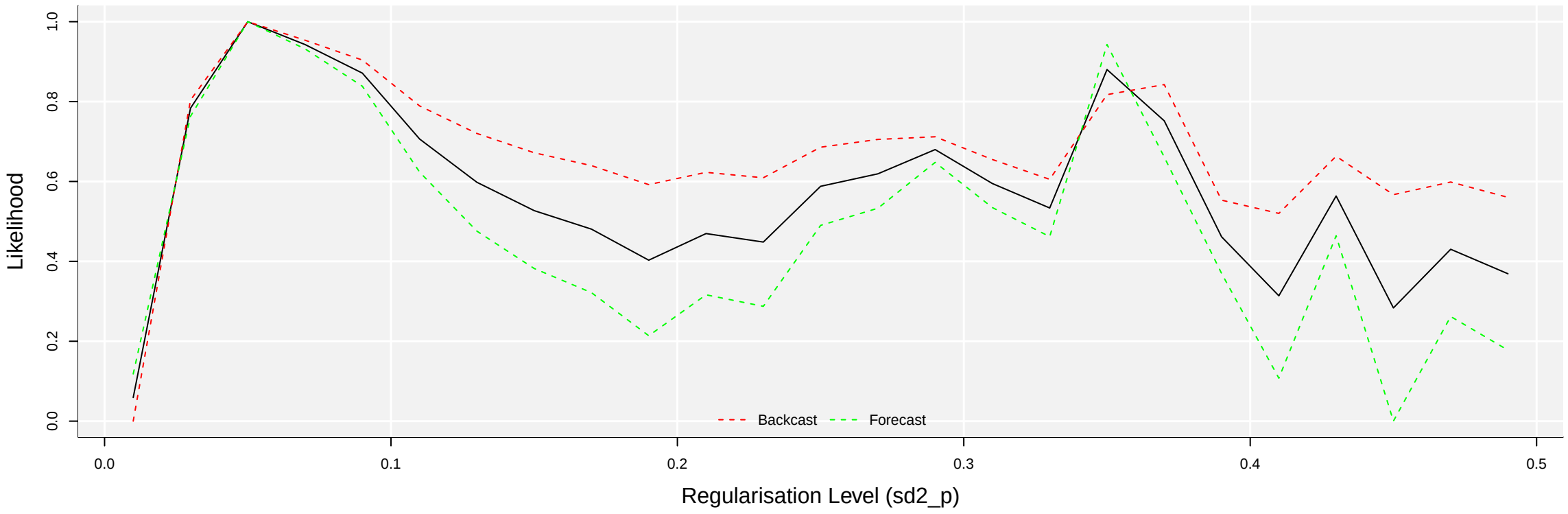

Histogram of residuals
Frequency
0
5
10
15
−3
−2
−1
0
1
2
residuals

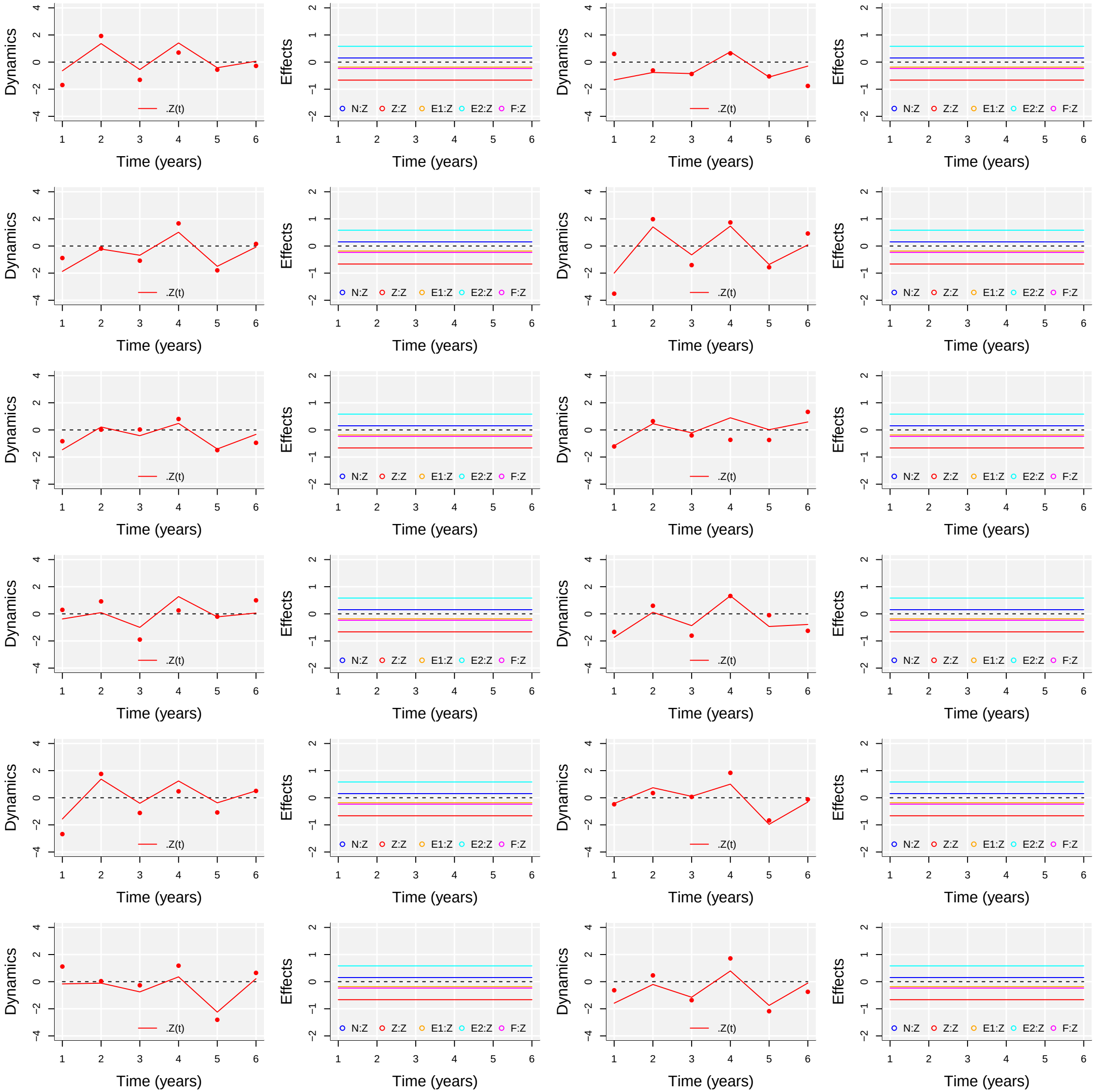

Log Body Size
Log Density
2.48
1.53
0.58
−0.38
−1.33
−2.29
−4.23
−3.01
−1.79
−0.57
0.65
1.87
3
2
2.5
1.5
1
0.5
0
−0.5
3.5
−1

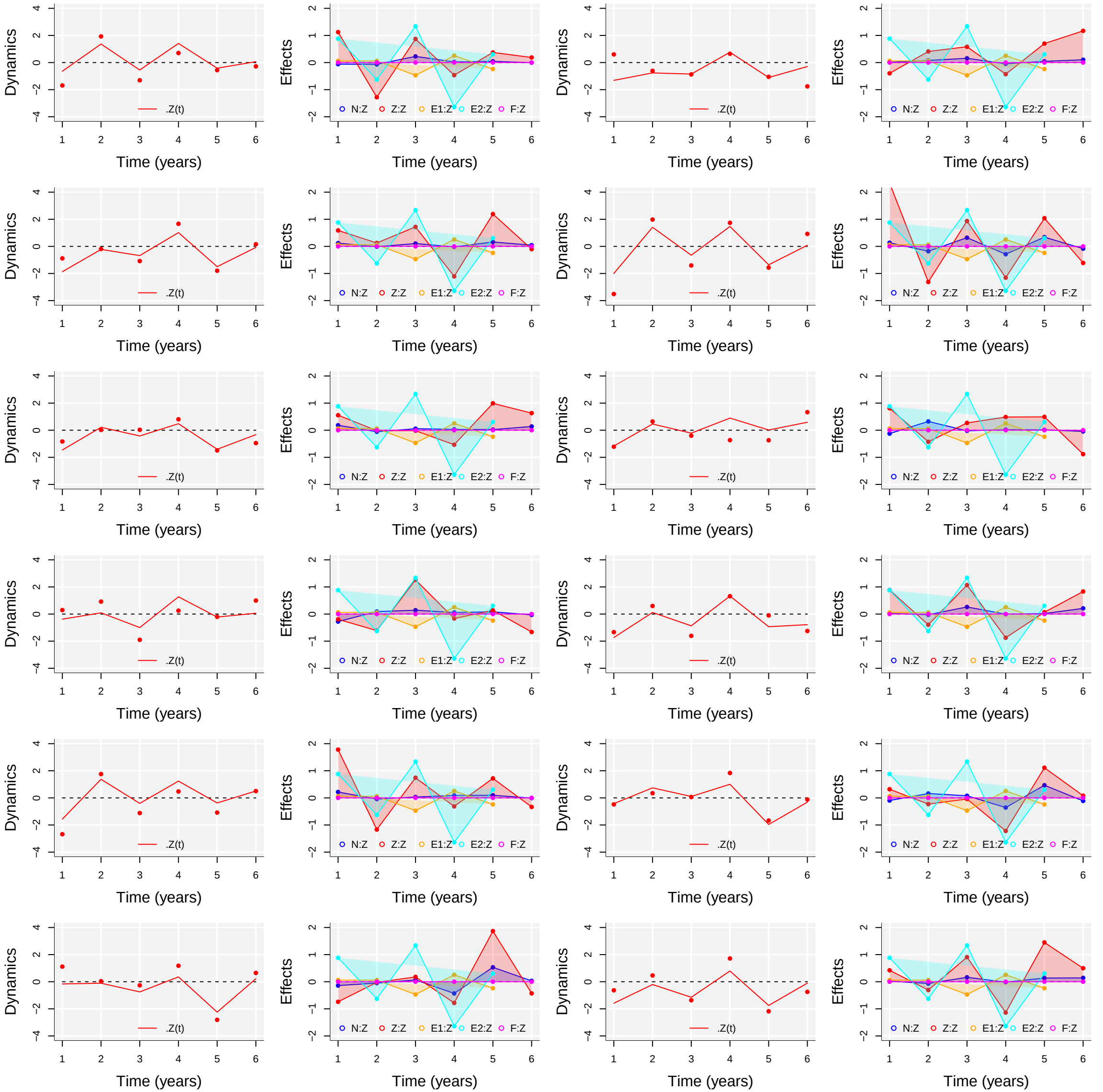

Log Body Size
Log Density
2.48
1.53
0.58
−0.38
−1.33
−2.29
−4.23
−3.01
−1.79
−0.57
0.65
1.87
2.5
2
1.5
1
0.5
0
−0.5
−1
3
−1.5

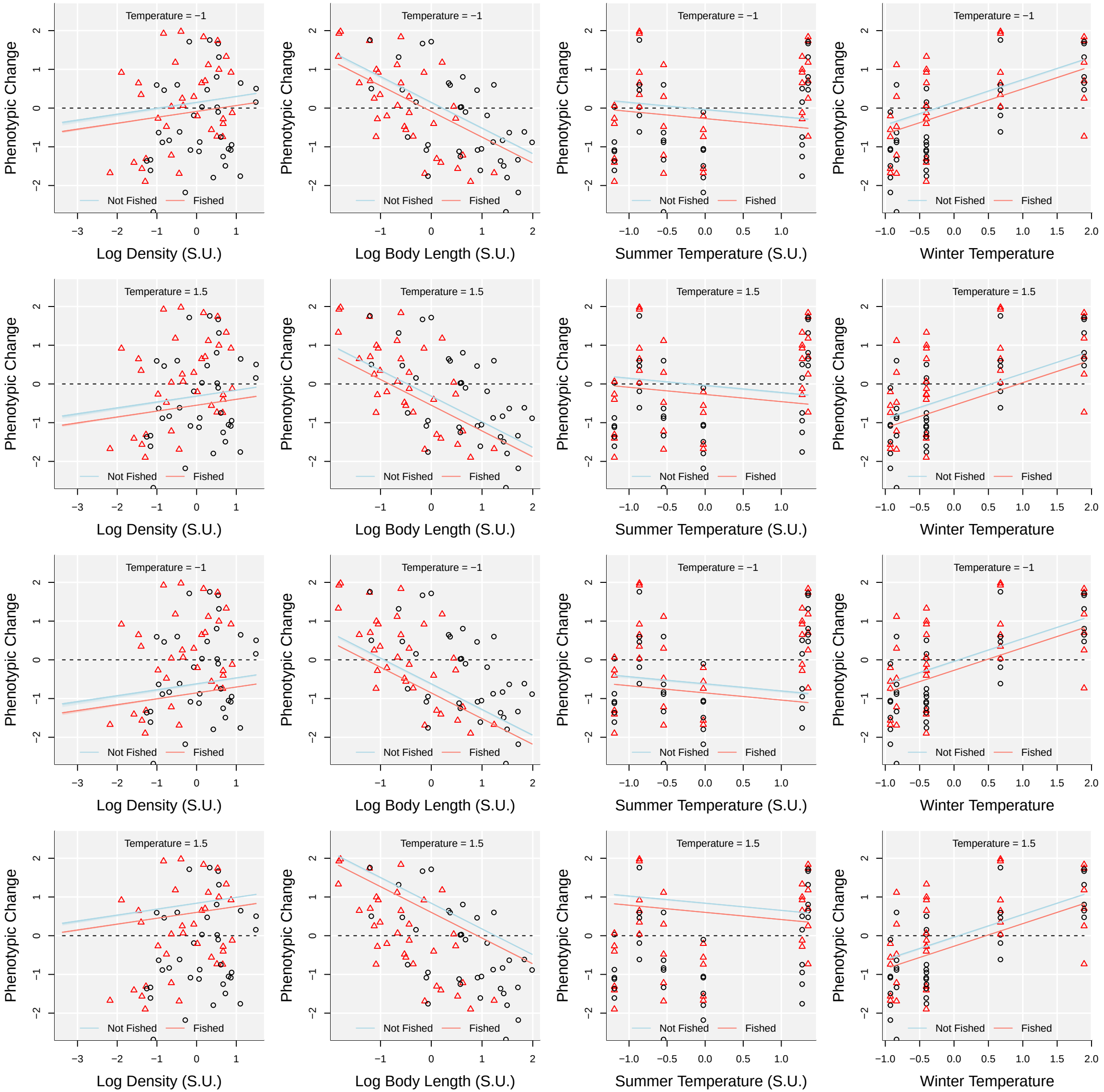

Backcast Likelihood
−2000
−2500
−3000
−3500
−4000
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

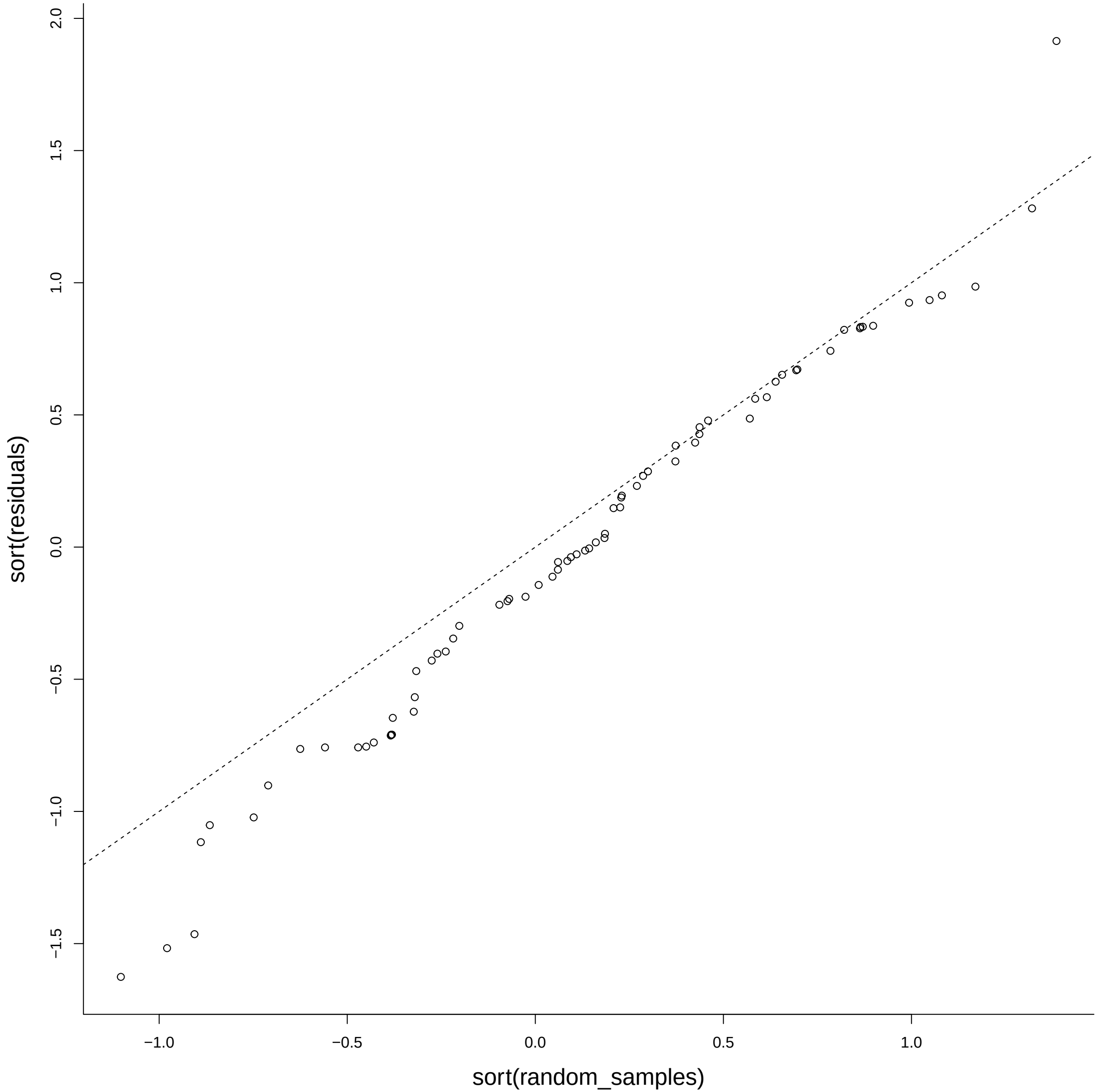

Forecast Likelihood
−2000
−3000
−4000
−5000
−6000
−7000
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

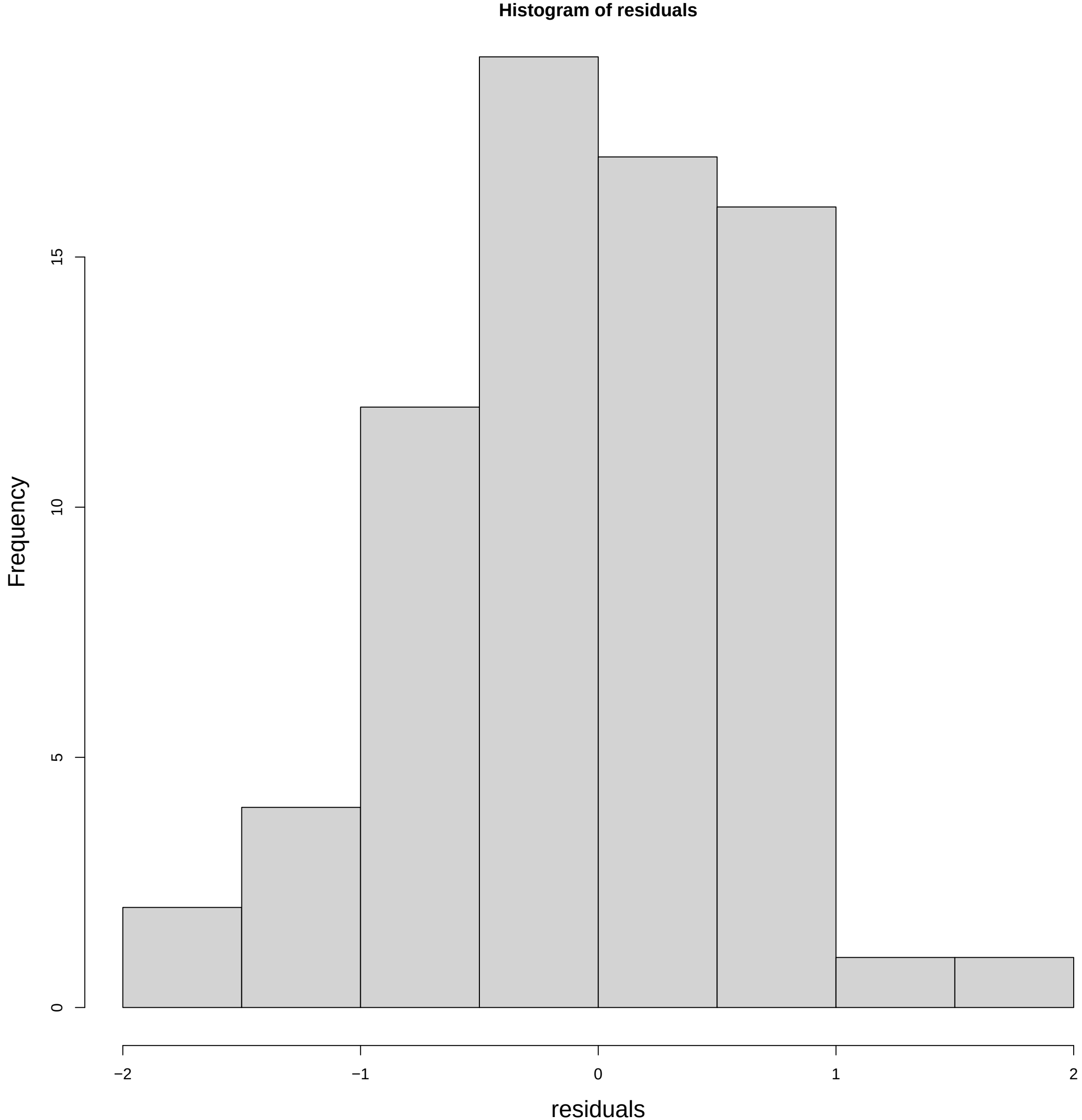

Likelihood
1.0
0.8
0.6
0.4
0.2
0.0
Backcast
Forecast
0.0
0.1
0.2
0.3
0.4
0.5
Regularisation Level (sd2_p)

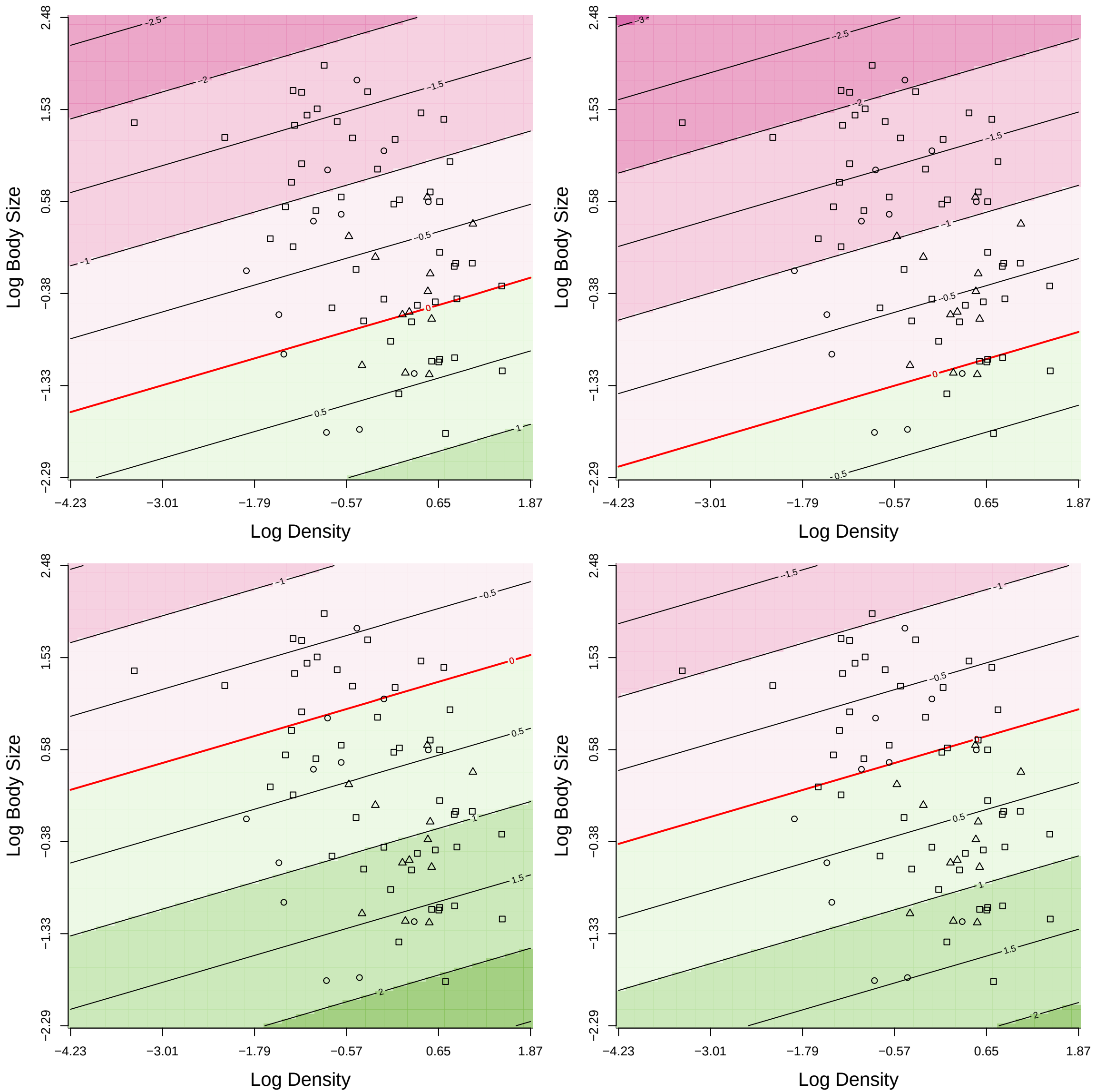

Dynamics
Effects
Time (years)
.Z(t)
N:Z
Z:Z
E1:Z
E2:Z
F:Z

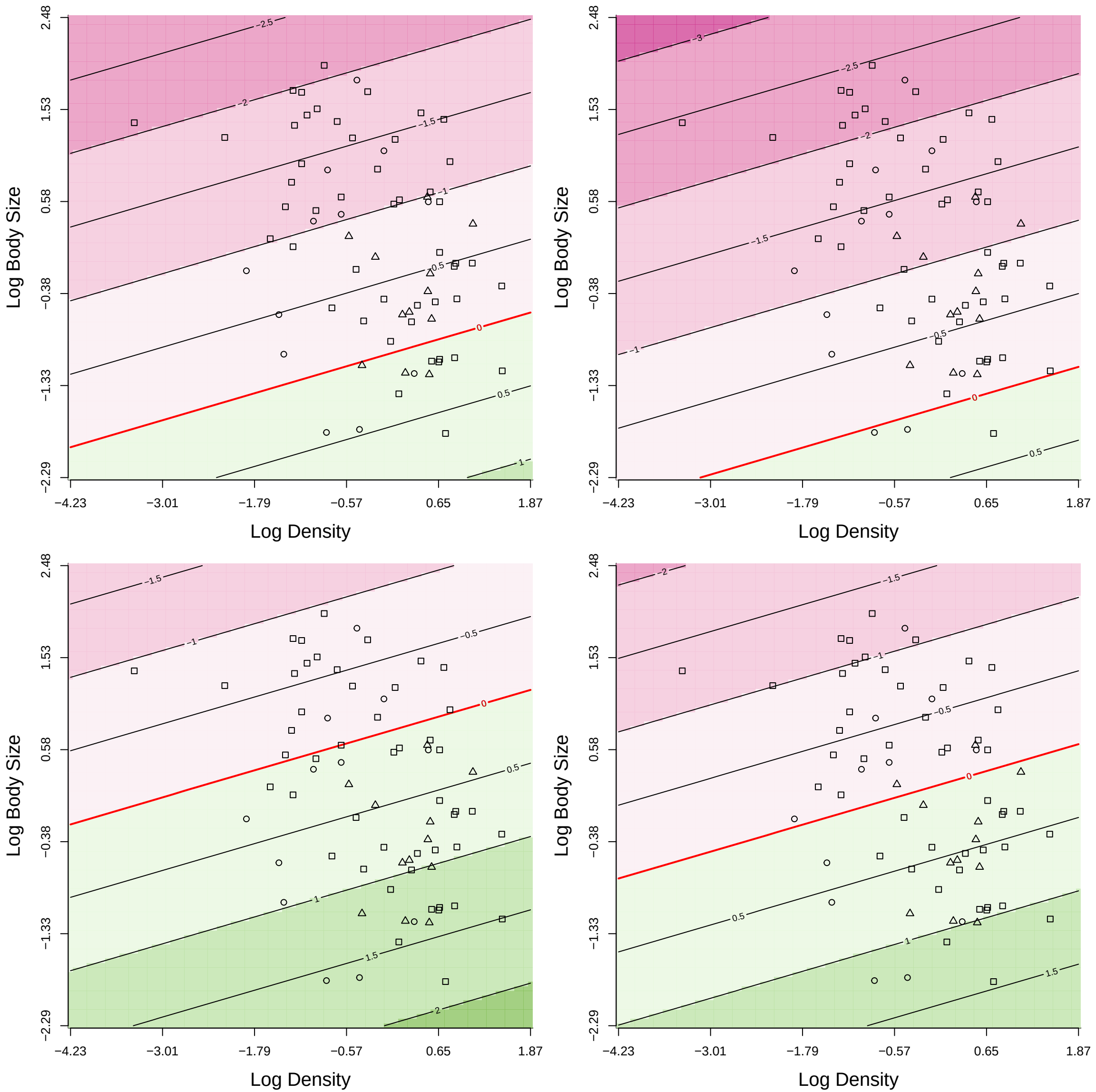

Dynamics
Effects
Time (years)
.Z(t)
N:Z
Z:Z
E1:Z
E2:Z
F:Z

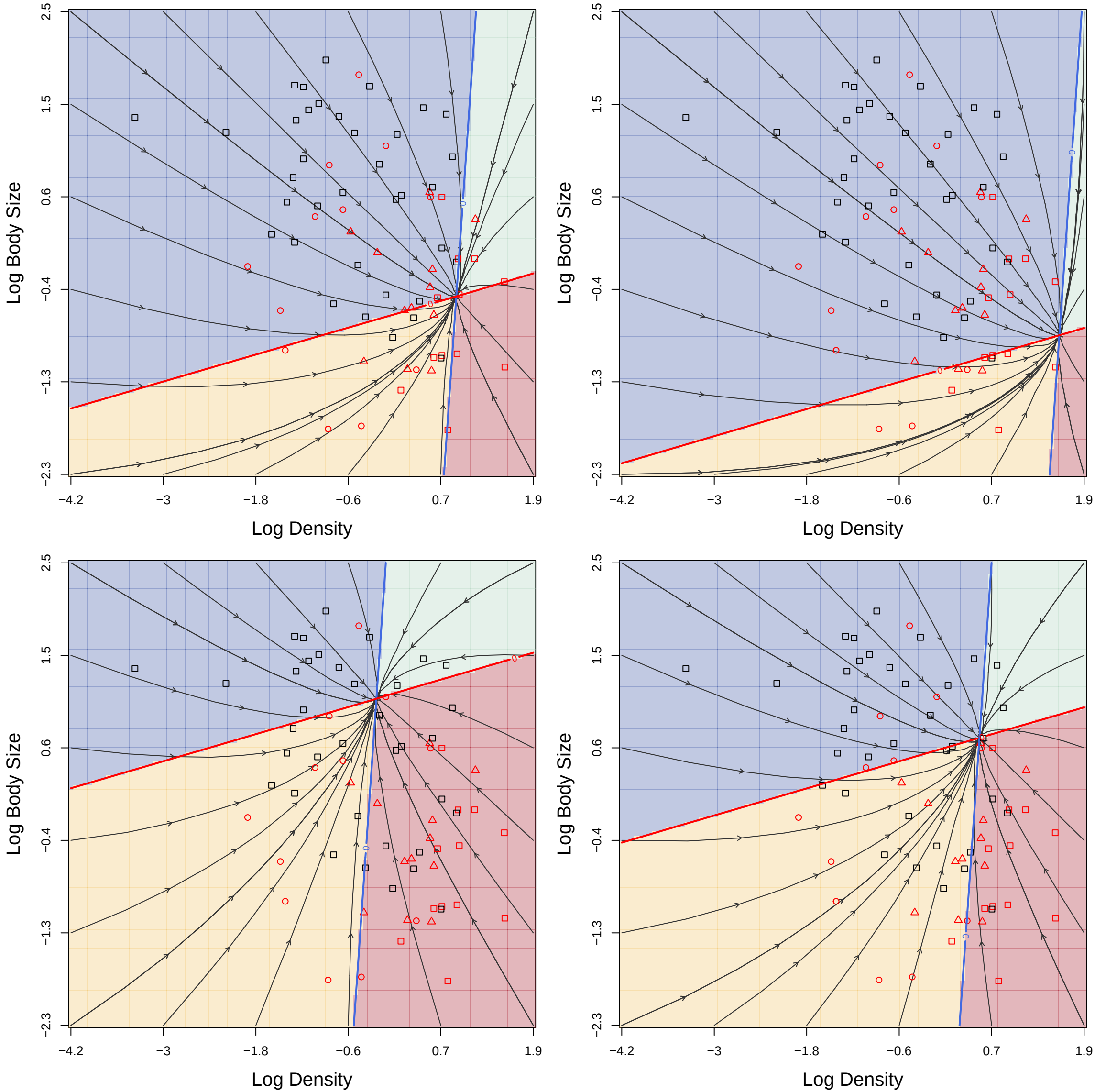

Temperature = −1
Temperature = 1.5
Phenotypic Change
Not Fished
Fished
Log Density (S.U.)
Log Body Length (S.U.)
Summer Temperature (S.U.)
Winter Temperature

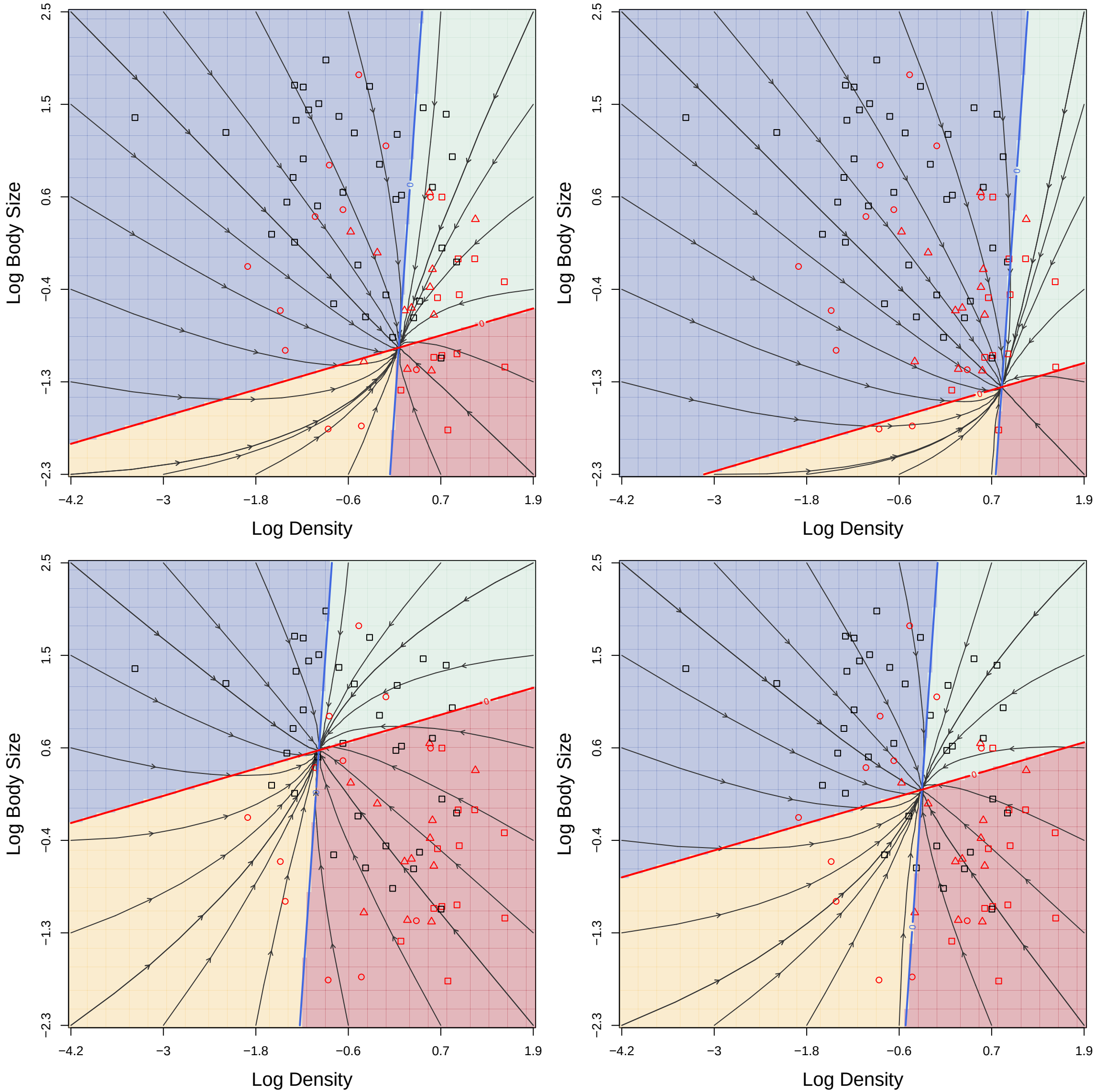

sort(residuals)
sort(random_samples)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0


Histogram of residuals
Frequency
0
5
10
15
−2
−1
0
1
2
residuals


Log Body Size
Log Density
2.48
1.53
0.58
−0.38
−1.33
−2.29
−4.23
−3.01
−1.79
−0.57
0.65
1.87
−3
−2.5
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2


Log Body Size
Log Density
2.48
1.53
0.58
−0.38
−1.33
−2.29
−4.23
−3.01
−1.79
−0.57
0.65
1.87
−2.5
−2
−1.5
−1
−0.5
0
0.5
1
−3
−2.5
−2
−1.5
−1
−0.5
0
0.5
−1.5
−1
−0.5
0
0.5
1
1.5
2
−2
−1.5
−1
−0.5
0
0.5
1
1.5


Log Body Size
Log Density
2.5
1.5
0.6
−0.4
−1.3
−2.3
−4.2
−3
−1.8
−0.6
0.7
1.9
0


Log Body Size
Log Density
2.5
1.5
0.6
−0.4
−1.3
−2.3
−4.2
−3
−1.8
−0.6
0.7
1.9
0